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M.Sc. Sem-2
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EXPERIMENT NO: 9 **DATE: / /**

AIM : DISTANCE MEASUREMENT USING SR-04 AND ESP32..

OBJECTIVE: The main objective of this experiment is,

A. To Interface an Ultrasonic sensor to measure distance

APPRATUS: Arduino IDE, ESP32, Breadboard, Resistors, Ultrasonic sensor SR-04, connecting wires, supply source, Display.

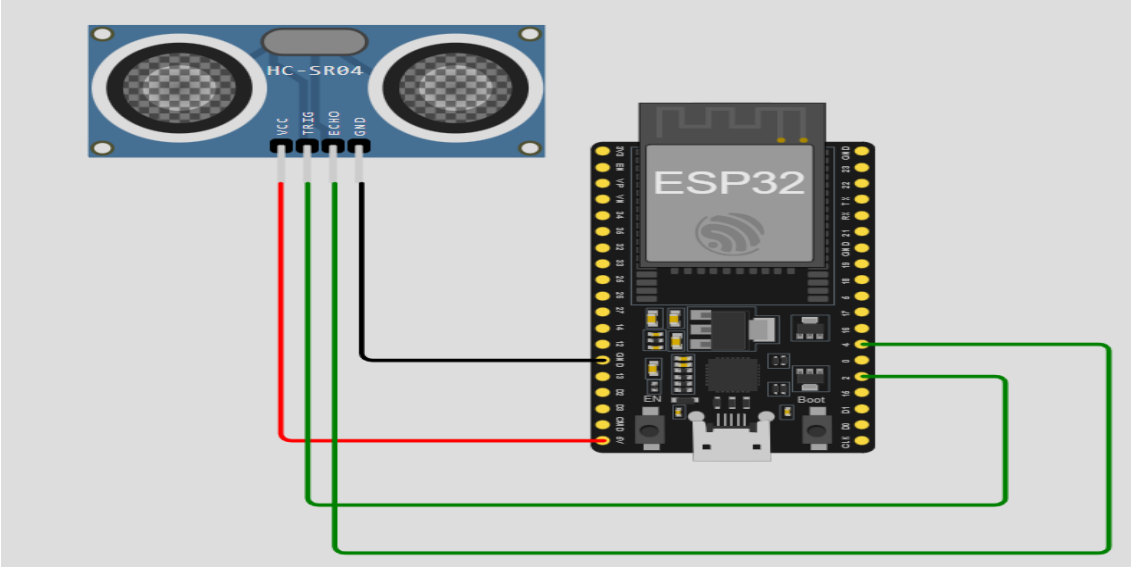
DESCRIPTION:

Ultrasonic sensors work by emitting high-frequency sound waves and measuring the time it takes for the sound to reflect off an object and return to the sensor. This "time of flight" is then used with the known speed of sound to accurately calculate the distance to the object, similar to echolocation used by bats and dolphins. The HC-SR04 ultrasonic sensor is a popular and affordable module that uses sonar to measure the distance to an object with high accuracy.

The HC-SR04 operates on the principle of echolocation (sonar):

1. The host microcontroller (e.g., Arduino) sends a short, 10µs high-level pulse to the Trig pin.
2. The sensor automatically transmits eight cycles of a 40 kHz ultrasonic sound burst and raises the Echo pin to a high state.
3. The sound waves travel outward, bounce off the target object, and the echo is detected by the receiver.
4. The Echo pin stays high for a time proportional to the travel duration of the sound waves.
5. The microcontroller measures this time (pulse width) and uses the speed of sound in air (approx. 340 m/s or 0.034 cm/µs) to calculate the distance using the formula: Distance = (Time × Speed of Sound) / 2. The division by 2 accounts for the sound's round trip

INTERFACING DIAGRAM:



PROCEDURE:

Give the connections to the ESP32 board as shown in the connection diagram.

- **Triggering:** The microcontroller (like an Arduino/ESP32) sends a **10µs HIGH pulse** to the sensor's **Trig** pin.
- **Sonic Burst:** Upon receiving this signal, the sensor automatically emits a burst of **eight ultrasonic pulses** at a frequency of **40 kHz**.
- **Echo Pin High:** Immediately after transmitting the burst, the sensor sets its **Echo pin to HIGH**.
- **Waiting for Echo:** The emitted sound waves travel through the air, hit an object, and bounce back toward the sensor.

- **Echo Pin Low:** When the sensor's receiver detects the returning sound waves, it pulls the **Echo pin LOW**.
- **Time Measurement:** The microcontroller measures the total time the Echo pin stayed HIGH. This duration represents the total **round-trip time** for the sound to travel to the object and back
- Reading will display on serial monitor

PROGRAM:

RESULTS:

CONCLUSION:

SAMPLE VIVA-VOCE QUESTIONS:

1) What is the purpose of the SR-04 ultrasonic sensor in this experiment?
2) What is the formula used to calculate the distance using the SR-04 ultrasonic sensor?
3) What is the purpose of the trigPin and echoPin connections in the SR-04 ultrasonic sensor?
4) How does the SR-04 ultrasonic sensor measure distance?
5) How is the SR-04 ultrasonic sensor connected to the ESP32?
6) What is the maximum range of the SR-04 ultrasonic sensor?
7) How can the accuracy of distance measurement be improved with the SR-04ultrasonic sensor?